

Routing Fundamentals: How Traffic Finds Its Way

Understand how Azure decides which path your network traffic takes — the GPS of your VNet

 **Beginner**

 **15 Minutes**

 **AZ-104 · Module 04**

Why This Matters

You've created a VNet with subnets and locked them down with NSGs. But how does traffic actually find its way from your web server to your database? Routing is the "GPS" of your network — it decides which path traffic takes. Understand routing, and you'll understand how your entire network communicates.

Before You Start

Prerequisites	VNets & Subnets and Network Security Groups
Time to understand	15 minutes
Difficulty	 Beginner (builds on NSGs, no advanced networking needed)
What you'll learn	How Azure routes traffic, system vs. custom routes, common routing patterns

The Simple Idea

What Is Routing?

Routing = Deciding the path that network traffic takes from source to destination.

Think of it like **GPS for data packets**.

```
Your data packet: "I need to go from 10.0.1.5 to 10.0.3.20"
      ↓
    Azure routing table
      ↓
    "I know a route to 10.0.3.0/24
    Go through subnet 10.0.3.0/24"
      ↓
    Data arrives at destination ✓
```

How Azure Decides Routes

Azure has a **routing table** (like a GPS navigation database) that says:

- "To reach 10.0.1.0/24, use local VNet"
- "To reach 10.0.2.0/24, use local VNet"
- "To reach 192.168.0.0/16, use VPN Gateway"
- "To reach anything else, use internet"

Two Types of Routes

Route Type	What It Does	Examples
System Routes	Built-in, automatic, managed by Azure	Internal VNet traffic, internet
Custom Routes (UDR)	You create them for special cases	Send traffic through firewall, VPN

How It Actually Works

Step 1: Traffic Arrives at Your Subnet

```
Data packet arrives:
From: 10.0.1.5 (WebServer)
```

To: 10.0.3.20 (Database)

Step 2: Subnet Looks at Its Routing Table

Destination	Next Hop	Purpose
10.0.0.0/16	Local VNet	Traffic within VNet stays local
10.0.1.0/24	Local Subnet	Traffic to this subnet is direct
10.0.2.0/24	Local Subnet	Traffic to app subnet is direct
10.0.3.0/24	Local Subnet	Traffic to DB subnet is direct
0.0.0.0/0	Internet	Everything else goes to internet

Step 3: Routing Lookup

Packet destination: 10.0.3.20

Azure checks routing table:

- └ Does it match 10.0.0.0/16? YES ✓
- └ Next hop: "Local VNet"
- └ Route: Send directly to destination within VNet

Result: Packet goes from 10.0.1.5 → 10.0.3.20 (direct path)

Step 4: NSG Check

Packet arrives at DB subnet:

- └ NSG checks inbound rule
 - "Is source 10.0.2.0/24 on port 1433?"
 - YES ✓
- └ NSG allows traffic

Result: Database receives the packet ✓

Mental Model: Routing as GPS Navigation

You're driving from point A to point B.

Your GPS (routing table):

- └ Route 1: "If destination is nearby, drive directly" (system route)
- └ Route 2: "If destination is far, take the highway" (system route)
- └ Route 3: "If you need to go through a checkpoint, use Route 66" (custom route)
- └ Route 4: "If no route exists, go to the main road" (default route)

Your data packet follows the same logic:

- └ System routes handle normal traffic (within VNet, to internet)
- └ Custom routes handle special cases (through firewall, VPN, etc.)

System Routes vs. Custom Routes

System Routes (Created Automatically)

Every VNet automatically gets these routes:

Destination	Next Hop	Created Automatically?
10.0.0.0/16	Local VNet	✓ Yes (VNet address space)
0.0.0.0/0	Internet	✓ Yes (default internet route)

Examples of what each means:

Route	Destination	Next Hop	When Used
Local VNet	10.0.0.0/16	Local	Traffic within your entire VNet
Internet	0.0.0.0/0	Internet	Traffic to anywhere not in your VNet

Custom Routes (UDRs - User-Defined Routes)

You create these for special cases:

Example UDR:

Destination: 192.168.0.0/16 (on-premises)

Next Hop: VPN Gateway

Purpose: Send traffic to on-premises through VPN

Why you'd create custom routes:

Use Case	Custom Route	Benefit
Send traffic through firewall	Destination: 0.0.0.0/0 → Next Hop: Firewall IP	All internet-bound traffic passes through firewall
Connect to on-premises	Destination: 192.168.0.0/16 → Next Hop: VPN Gateway	Corporate network access
Send to security appliance	Destination: 10.0.0.0/16 → Next Hop: NVA (network appliance)	All internal traffic inspected

Worked Example: Real Scenario

The Scenario

TechCorp's network:

- 3 subnets within VNet: Web (10.0.1.0/24), App (10.0.2.0/24), DB (10.0.3.0/24)
- Azure Firewall in hub VNet protecting everything
- VPN connection to on-premises network (192.168.0.0/16)

Goal: Route traffic correctly through firewall and VPN.

Step 1: Default System Routes (Automatic)

App Subnet (10.0.2.0/24) Routing Table (Default):

Destination	Next Hop	Priority
10.0.0.0/16	Local VNet	1 (All internal VNet traffic)
0.0.0.0/0	Internet	2 (Everything else to internet)

AppServer (10.0.2.10) wants to send data:

Case 1: To Database (10.0.3.20)
└ Destination: 10.0.3.20

```
└─ Match: 10.0.0.0/16? YES ✓  
└─ Route: Local VNet (direct path)  
└─ Result: 10.0.2.10 → 10.0.3.20 directly ✓
```

Case 2: To Internet

```
└─ Destination: 8.8.8.8 (Google)  
└─ Match: 10.0.0.0/16? NO  
└─ Match: 0.0.0.0/0? YES ✓  
└─ Route: Internet (default route)  
└─ Result: 10.0.2.10 → Internet directly ✓ (No firewall!)
```

Problem: Traffic to internet bypasses firewall. We need custom routes.

Step 2: Add Custom Routes (Force Through Firewall)

Admin creates custom route:

Destination: 0.0.0.0/0 (all internet traffic)

Next Hop: Azure Firewall (10.0.0.4)

Priority: Higher than default

App Subnet (10.0.2.0/24) Routing Table (Updated):

Destination	Next Hop	Priority
0.0.0.0/0	Firewall IP	1 (Custom route - checked first!)
10.0.0.0/16	Local VNet	2
0.0.0.0/0	Internet	3 (Default - not used now)

AppServer (10.0.2.10) wants to send data:

Case 1: To Database (10.0.3.20)

```
└─ Destination: 10.0.3.20  
└─ Match: 0.0.0.0/0? YES, but next hop is 10.0.0.0/16  
└─ No wait, check priority 2 first  
└─ Match: 10.0.0.0/16? YES ✓  
└─ Route: Local VNet (direct, bypass firewall—OK for internal)  
└─ Result: 10.0.2.10 → 10.0.3.20 directly ✓
```

Case 2: To Internet

```
└─ Destination: 8.8.8.8 (Google)  
└─ Match: 0.0.0.0/0? YES ✓ (custom route, high priority)
```

```
└ Route: Firewall IP (10.0.0.4)
└ Result: 10.0.2.10 → Firewall → Internet ✓ (Firewall inspects!)
```

Result: Custom route forces internet-bound traffic through firewall.

Step 3: Add VPN Route (For On-Premises Access)

Admin creates another custom route:

Destination: 192.168.0.0/16 (on-premises)

Next Hop: VPN Gateway

Priority: 1

App Subnet (10.0.2.0/24) Routing Table (Final):

Destination	Next Hop	Priority
192.168.0.0/16	VPN Gateway	1 (On-prem traffic via VPN)
0.0.0.0/0	Firewall IP	2 (Internet via firewall)
10.0.0.0/16	Local VNet	3 (Internal via VNet)
0.0.0.0/0	Internet	4 (Default - not reached)

AppServer (10.0.2.10) wants to send data:

Case 1: To Database (10.0.3.20)

```
└ Match: 192.168.0.0/16? NO
└ Match: 0.0.0.0/0? YES, but check next priority
└ Match: 10.0.0.0/16? YES ✓
└ Route: Local VNet
└ Result: Direct to database ✓
```

Case 2: To On-Premises Server (192.168.1.50)

```
└ Match: 192.168.0.0/16? YES ✓
└ Route: VPN Gateway
└ Result: 10.0.2.10 → VPN Gateway → VPN Tunnel → On-premises ✓
```

Case 3: To Internet (8.8.8.8)

```
└ Match: 192.168.0.0/16? NO
└ Match: 0.0.0.0/0? YES ✓
└ Route: Firewall IP
└ Result: 10.0.2.10 → Firewall → Internet ✓
```

Common Mistakes (What NOT to Do)

✗ Mistake 1: Unexpected Default Route

Wrong

Admin creates custom route:

Destination: 0.0.0.0/0 → VPN Gateway

All traffic goes to VPN, including:

- └ Internal traffic (should stay local) ✗
- └ Internet traffic (should go to internet or firewall) ✗
- └ Database traffic (should be direct) ✗

Result: Everything bottlenecks through VPN

Fix

Be more specific:

- └ 192.168.0.0/16 → VPN Gateway (only on-prem traffic)
- └ 10.0.0.0/16 → Local VNet (internal traffic)
- └ 0.0.0.0/0 → Internet or Firewall (external traffic)

Each traffic type follows the correct path ✓

Why it fails: Too broad a route causes unnecessary traffic redirection.

✗ Mistake 2: Routing Loop

Wrong

Custom route 1: 0.0.0.0/0 → Firewall (10.0.0.4)

Custom route 2: 0.0.0.4/32 → Internet

Packet goes: App → Firewall → Internet

But what if firewall tries to reach internet?

- └ Matches 0.0.0.0/0 → Firewall again
- └ Infinite loop! ✗

Fix

Firewall routes to internet directly (exclude from custom route):

- └ Custom route: 0.0.0.0/0 → Firewall
- └ Except: Firewall's NIC route: 0.0.0.0/0 → Internet (direct)

Packet goes: App → Firewall → Internet (no loop) ✓

Why it fails: Packet gets stuck in a loop.

✗ Mistake 3: Too Many Custom Routes

Wrong

Create 100 custom routes for every possible destination

- └ 8.8.8.8/32 → Internet
- └ 1.1.1.1/32 → Internet
- └ ... (95 more individual IPs)
- └ 192.168.0.0/16 → VPN

Result:

- └ Difficult to maintain
- └ Error-prone
- └ Confusing for other admins
- └ Harder to troubleshoot

Fix

Use aggregate routes (CIDR notation):

- └ 0.0.0.0/0 → Firewall (one route handles everything)
- └ 192.168.0.0/16 → VPN (one route for all on-prem)

Much simpler and easier to manage ✓

Why it fails: Complexity = mistakes and maintenance headaches.

Routing Decision Order (Route Priority)

When traffic arrives, Azure checks routes in this order:

1. Most Specific Match (Longest Prefix)

Destination: 10.0.2.5

Routes available:

- └─ 10.0.2.0/25 (specific)
- └─ 10.0.2.0/24 (less specific)
- └─ 10.0.0.0/16 (even less specific)

Azure picks: 10.0.2.0/25 (most specific) ✓

2. Custom Routes Before System Routes

If both custom and system routes exist for same destination:

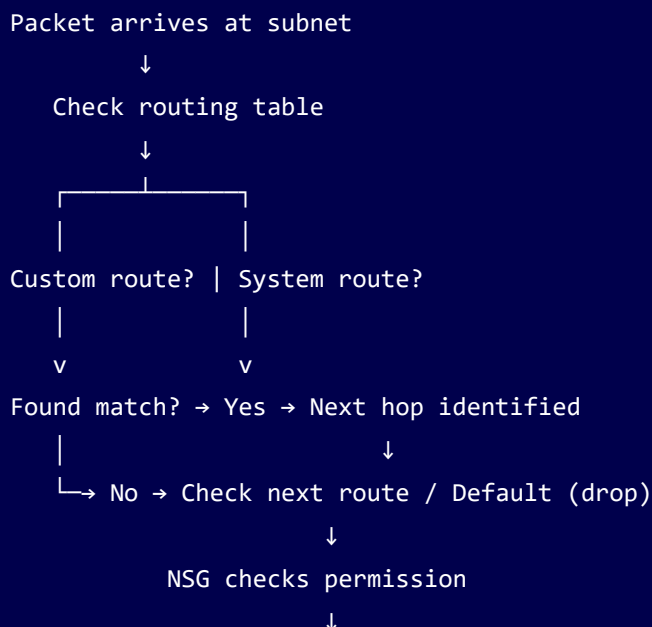
- └─ Custom route wins

3. Default Deny If No Match

If no route matches:

- └─ Traffic is dropped (denied) X

Routing Decision Flowchart





Summary: Three Layers of Network Control

Layer	Tool	Controls	Example
1. Routing	Routing Tables (System + UDR)	Which path traffic takes	Route all internet traffic through firewall
2. Security	Network Security Groups (NSG)	Which traffic is allowed	Allow port 443 from web subnet to app subnet
3. Access	RBAC	Who can change routing/NSGs	Only network admins can modify routing tables

All three together provide complete network control.

How This Connects to Other Topics

Related to Module 01 (Identity & Governance)

- **RBAC + Routing:** Use RBAC to control who can modify routing tables
- **Policy + Routing:** Enforce routing policies (e.g., all traffic must go through firewall)

Related to Module 02 (Storage)

- **Storage Firewall Routes:** Combined with routing, restrict storage access to specific networks
- **Service Endpoints:** Routes direct traffic to storage without public internet

Related to Module 03 (Compute)

- **VM to VM Communication:** VMs in same VNet route directly (system route)

- **VM to Database:** Routing determines path through NSG and firewall

Related to Module 05 (Monitor)

- **Route Metrics:** Monitor traffic flow through different routes
- **Network Watcher:** Trace routing paths to diagnose connectivity issues

See It In Action

Associated lab: Lab 14: Configure Routing & UDRs

Suggested learning sequence:

1.  Read **VNets & Subnets**
2.  Read **Network Security Groups**
3.  Read this doc (Routing Fundamentals)
4.  Work through Lab 14 (hands-on routing configuration)
5.  Read **VNet Peering** (routing between peered VNets)

Key Takeaways



Summary

- **Routing = GPS for data packets** (decides which path traffic takes)
- **System routes are automatic** (internal traffic, internet access)
- **Custom routes (UDRs) are for special cases** (firewall, VPN, appliances)
- **More specific route wins** (longer CIDR prefix = higher priority)
- **Custom routes override system routes** (same destination priority)
- **Traffic flows based on routing table** then **checked by NSG rules**
- **Routing + NSG = complete traffic control** (routing decides path, NSG decides permission)

Next Steps

1. **Learn:** Read this doc (you're here)
2. **Practice:** [Lab 14: Configure Routing & UDRs](#)
3. **Connect:** Read [VNet Peering](#) (routing between peered VNets)
4. **Advance:** Read [VPN & ExpressRoute](#) (routing to on-premises)
5. **Secure:** Advanced — Azure Firewall with routing rules

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